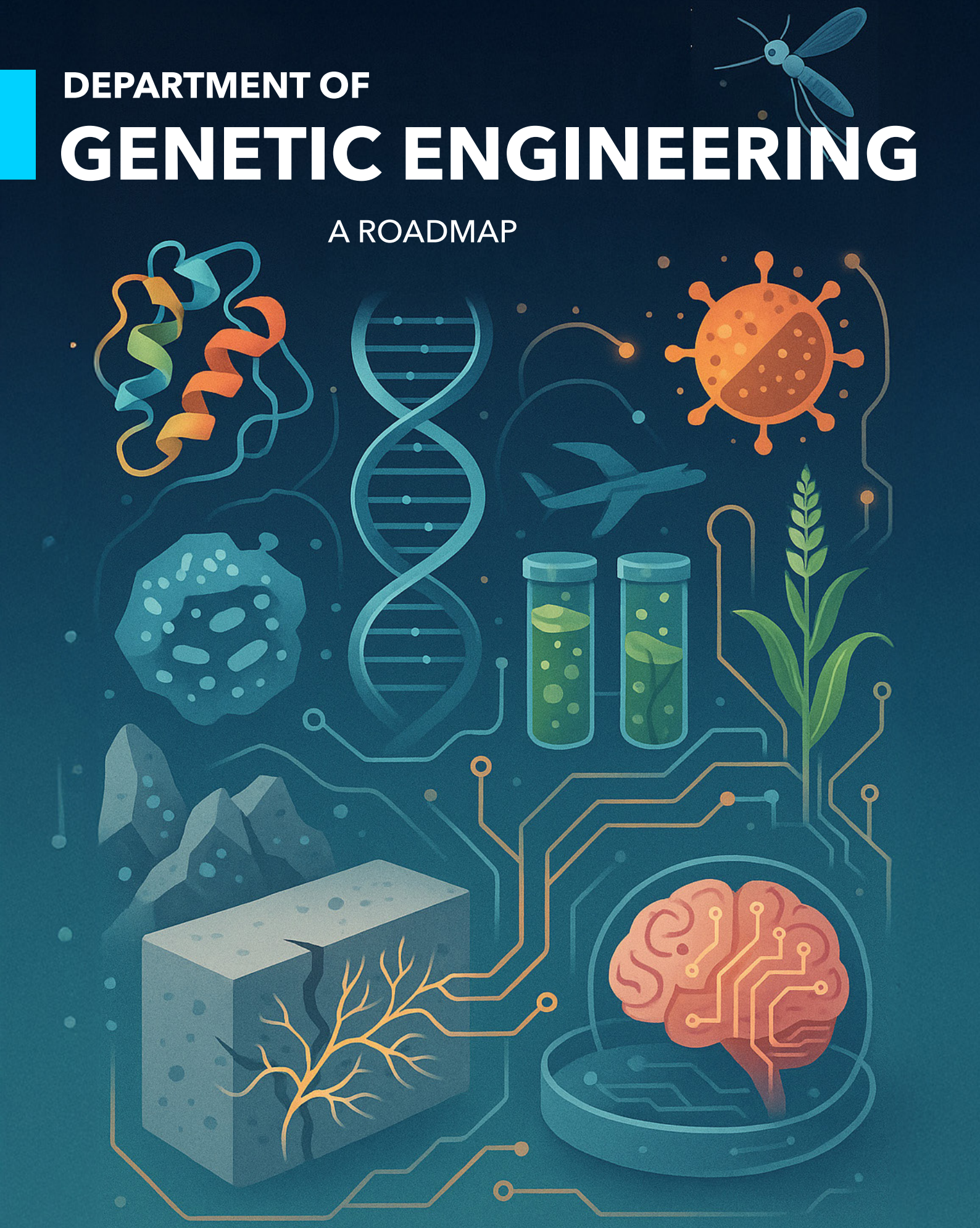
DEPARTMENT OF GENETIC ENGINEERING

A ROADMAP



NIKO MCCARTY & CHRISTOPHER A. VOIGT

The Vision

The design of life is the zenith of engineering disciplines. Over the last half-century, researchers have made rapid advances in creating DNA that reprograms living cells. Biotechnology now spans all aspects of society, from living medicines to new sources of food, chemicals, and minerals. Yet, as a discipline, genetic engineering emerged without an intellectual home, siloed across departments. Projects are housed in departments due to tradition; for example, corn engineering in Agrosciences, microbial biomanufacturing in Chemical Engineering, and human T-cell design in Biomedical Engineering. Legacy coursework and faculty compositions draw students away from core competencies in genetic engineering. As a result, students receive only partial educations, which limits their career options and forces them to pursue long PhDs and postdoctoral training for positions where it should not be required.

A focused Department of Genetic Engineering will train future students in the theory, practice, and ethics of the field. The workforce must be prepared for both modern and yet-to-be-envisioned applications, such as advanced materials, mining, and infrastructure. Students need to become capable designers who understand the principles and mathematics behind reprograming every kingdom of life.

Department Mission

Advance the practice and applications of genetic engineering to benefit society. Foster innovation and excellence in research and education, equipping students with the skills and ethical grounding necessary to contribute responsibly.

Core pedagogical philosophies

1

Universal design across the kingdoms of life.

Genetic engineers should be able to write genetic code for all biological systems, spanning viruses and bacteria to plants and animals. Differences in basic biology driving engineering decisions need to be understood.

2

Mathematics-guided predictive engineering.

Genetic engineering is a predictive and quantitative science based on physical principles. Biophysics, and related computational tools, will be applied to the design of living systems. New theory is required as biological systems exist at the boundary of physical regimes.

3

Precision and comprehensive measurements.

To effectively engineer a system, one must see the impact of a genetic change. Illuminating biological systems requires the integration of measurement methods across scales in time and space



Pharmaceuticals

"Everybody is moving into quantitative realms, and it is difficult to hire really good […] genetic engineers who know how to think about living systems in a quantitative way."

John Davis, Novo Nordisk



Defense

"There is always a need for people to understand how to design [materials]. What can we learn about how organisms make materials? Artificial intelligence and genetic engineering have gone hand-in-hand from the beginning and will continue to be powerful."

Melissa Rhoads, Lockheed Martin



Agriculture

"[Syngenta] would hire genetic engineers from this program. Undergraduate applicants have been mainly in biology or chemistry. We have not found an undergraduate with the skills we are looking for."

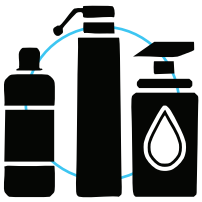
Lauren Ray, Syngenta



Mining

"We are doing genetic engineering [...] to access ore that were not viable before to be extracted. Rio Tinto is so new to genetic engineering for biomining, but they think this will be the next big frontier."

Kaci Jenkins, Rio Tinto



Consumer Products

"Industry right now wants engineers because they are good at problem solving in a way that most biological scientists are not. Purely biological people often never learn how to build and test and iterate."

Rob Markowitz, Estee Lauder



Life Sciences

"I love the idea of having a curriculum for genetic engineering. It will be super valuable and the students coming through would be very employable in the private sector."

Jonathan Chestnut, Thermo Fisher

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Executive Summary

Progress in genetic engineering was initially stunted by a lack of design tools, physical difficulties in DNA and strain construction, and an inability to visualize the impact of genetic changes. The field was dominated by the idiosyncrasies of working with different organisms. Now, with tools spanning AI-guided protein design to genome editing and low-cost -omics, the tools to engineer living cells have unified, thus preparing the field to become a true engineering discipline.

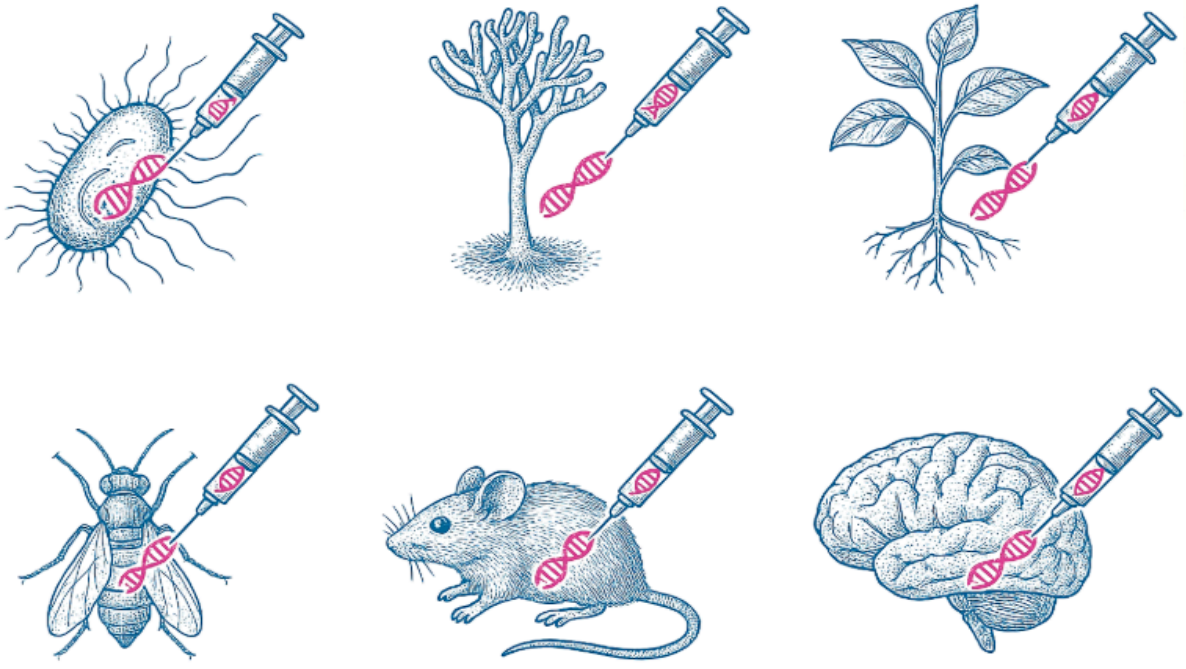
The early struggles with genetic engineering in the 1970s led to a siloing of organisms and applications across departments. This partitioning squeezed new genetic engineering coursework into curricula that had existed for a century. Today, most degree programs devote few classes exclusively to genetic engineering. In analyzing 90 undergraduate programs, genetic engineering coursework was found to be sparse, ranging from 3-20% (0-3 classes). Consequently, after a typical 4-year bachelor's degree, graduates lack the foundational skills to work as genetic engineers. Thus, PhD and postdoctoral training is required, slowing their entry into the workforce.

This roadmap lays out a new Department of Genetic Engineering as a top-tier, PhD-granting research and education program housed at a university. It will assemble a vibrant community focused on programming biological systems for current and yet-to-be-envisioned applications. The unifying pedagogical principle is the application of mathematics to genetic design and a core courseload is proposed. The focus in not on today's techniques. Instead, students will master the fundamentals needed to use and develop tomorrow's design tools and engineering methods, with broad coverage of bacterial, fungal, plant, and animal engineering. Ethical, legal, social implications (ELSI) of genetic engineering

will be integrated into both undergraduate and graduate curricula.

Upon completion of their degree, graduating undergraduate will be prepared to enter industry and apply their art. Engagement with industry will ensure that the coursework remains relevant. To that end, we interviewed employees from 108 companies and national labs representative of diverse sectors. Graduates with a BSE could also enter PhD programs for scientist track positions, start labs in academia, or pursue medical, legal, governmental, or other careers.

Faculty will be drawn from diverse departments and areas of expertise. Housing such a program will require a new approach to building design. Architectural plans envision flexible space, designed to intermix researchers. Faculty can reconfigure these spaces to house the large and collaborative projects underlying modern biology. Additionally, the building will enable public engagement through museum-like displays and open sightlines into research spaces, ensuring transparency about the work being done and its broader implications. Because genetic engineering is one of the most controversial disciplines in society, the department will be a critical nexus for evaluating and communicating its benefits, risks, and impacts.



Emergence of Genetic Engineering

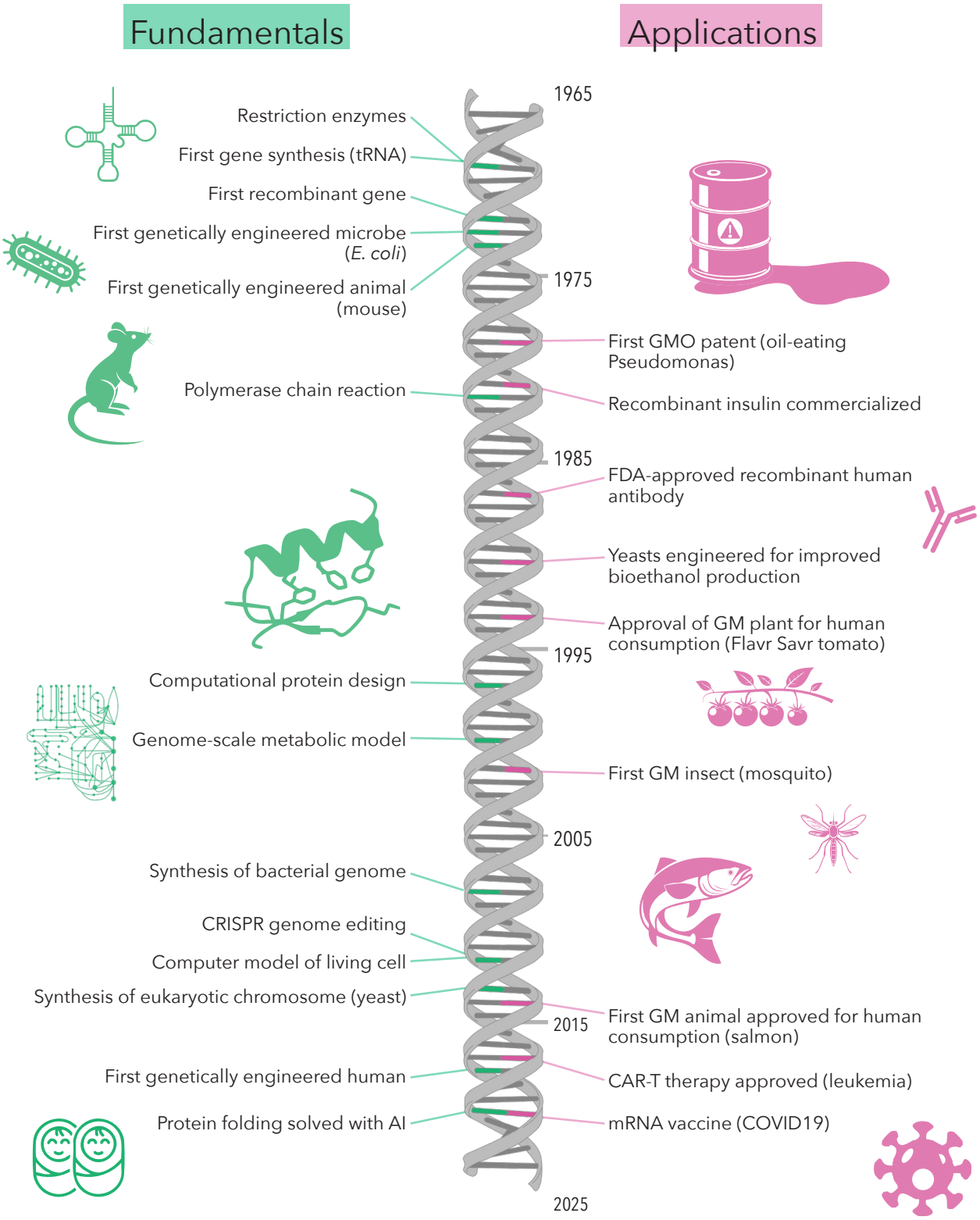


The first genetically modified organisms emerged in the 1970-80s. In 1973, Herbert Boyer and Stanley Cohen moved the human insulin gene into the bacterium *Escherichia coli*, which became an FDA-approved drug. In 1974, the first genetically-modified animal (a mouse) was made. By 1982, the first transgenic plant (tobacco) emerged. These achievements spurred debates on ethics and safety, epitomized by the Asilomar Conference of 1975.

Early genetic engineering required physically isolating DNA from one organism and transplanting it to another. Tools like PCR and restriction enzymes streamlined this process by enabling DNA amplification, precise cutting, and insertion into new hosts.

Viewing cells as factories, Metabolic Engineering sought to improve the production of chemicals, fuels, and pharmaceuticals. Chemical Engineers pioneered this concept in the early 1990s, leveraging kinetics, transport, and thermodynamics to optimize the carbon flux from sugar, through metabolism, to the desired chemical product. These researchers had a connection to the chemical products that had previously been sourced from petroleum. Subsequently, Synthetic Biology began in the early 2000s to broaden the engineering principles applied to build cells. The field was powered by new capabilities in the synthetic construction of DNA, genome editing, and rapid expansion of DNA sequence databases. Increasingly, computer aided design (CAD) tools became routine for the quantitative design of DNA. Further, species barriers began to break down with manipulation tools agnostic to organism type.

Timeline of Genentic Engineering

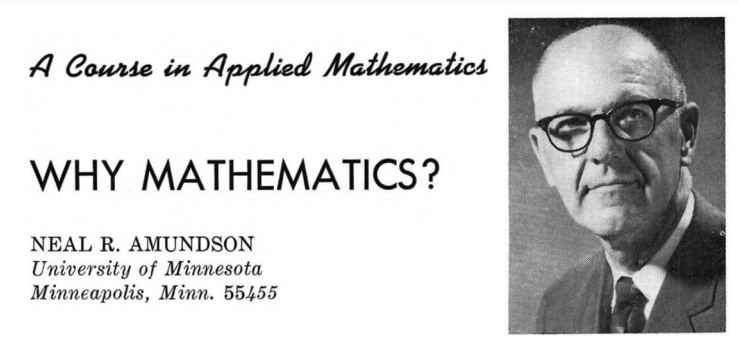


A Revolution in Chemical Engineering

Case Study

In 1859, crude oil was discovered in the United States and by the 1880s the annual production was 30 million barrels. Beyond fuel, it led to a surge of organic chemicals, prompting the first Chemical Engineering department at MIT in 1888, spreading to Europe (1905) and Asia (1921). In these early years, the research varied widely and this was reflected in the undergraduate programs. Classes were empirical, with no focus on underlying physiochemical principles. Graduates found themselves unprepared for cutting-edge projects in the rapidly-growing oil and chemicals industry.

Neal Amundson, a ChemE undergraduate at the University of Minnesota, found his coursework "mostly experimental and dull." This changed in 1944 during a PhD-year visit to Brown University, a renowned center for applied mathematics. There, Amundson recognized that mathematics provides a link to physiochemical phenomena that could lead to disruptive technologies. This convergence became the backbone of modern chemical engineering, as mathematics expanded the ability to study and design systems beyond what could be characterized experimentally. Back at Minnesota, Amundson launched a popular "Mathematics for Chemical Engineers" class and eventually became department head.


A Course in Applied Mathematics

WHY MATHEMATICS?

NEAL R. AMUNDSON
University of Minnesota
Minneapolis, Minn. 55455

As department head, Amundson was supported by an unorthodox Dean. Together, they initiated an era of "hiring sprees" that rapidly brought in faculty with diverse skills. In fact, they avoided hiring Chemical Engineers to avoid dogma and instead cast a wide net, even hiring a microbiologist. Through nationwide collaboration, the pedagogical triad of chemical engineering was founded: thermodynamics, kinetics, and transport phenomena. Faculty began to have additional defined roles, partly defined by the curriculum, such as Process Control, Catalysis, and Biotechnology. This structure spread globally with many programs using the same course structure and textbooks, much of which is still in place today. Collectively, these developments in the field led to the toolset of Chemical Engineering to be applied to many interdisciplinary problems, inconceivable in the 1950s.

Kenneth H. Keller, Quality and Community: The Story of Minnesota's Department of Chemical Engineering and Materials Science, 2022, The Reagents of the University of Minnesota.

H. Ted Davis, The Making of a Fine Department, Remarks to Guests of President Nils Hasselmo, Eastcliff 21 February 1995

Genetic Engineering Mathematics

Intercellular transport

Sensing and Regulation

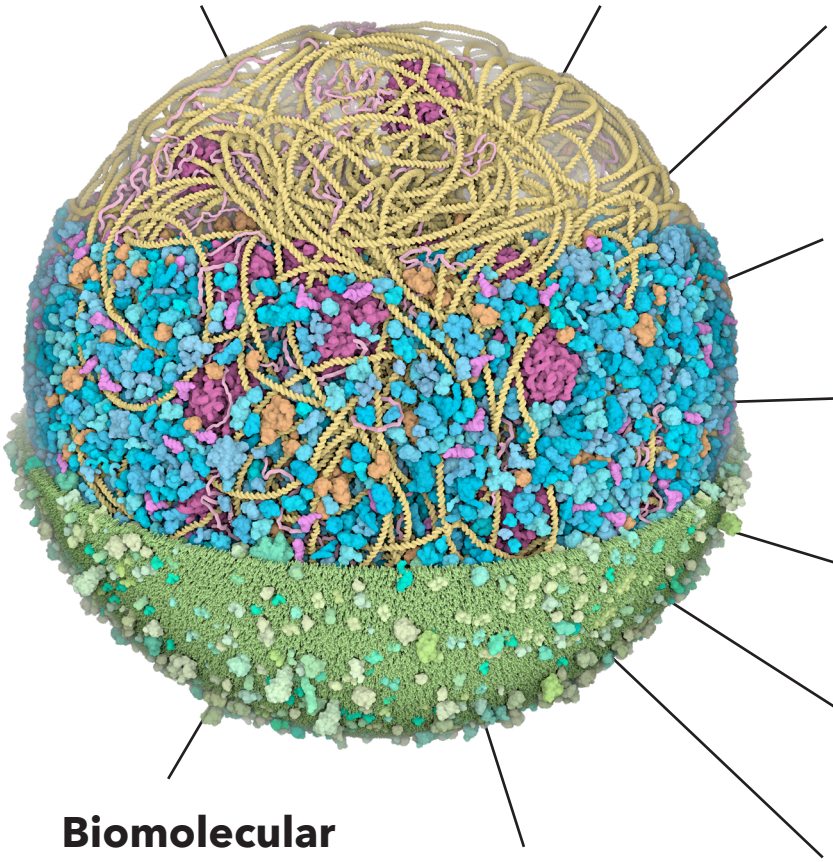
- Genetic circuit design automation
- Control theory

Transcription/Translation

- Non-equilibrium thermodynamics
- Transport phenomena

Metabolism

- Linear algebra
- Genome-scale metabolic model
- Retrosynthesis algorithms



Organisms

Diffusion

- Brownian ratchets
- Molecular motors
- Electrostatics

Multicellular simulations

- Physiological models
- Growth models
- Agent-based modeling

Whole-cell simulations

- AI/ML
- Advection-diffusion
- Stochastic simulations

Membranes

- Self-assembly

Biomolecular

Protein design and folding

- Thermodynamics
- AI/ML
- Rosetta/Alphafold
- Molecular dynamics

Metabolites

- Kinetics
- Density functional theory

RNA/DNA folding

- Statistical mechanics
- mfold

Image by Martina Maritan, Scripps Research. Illustrations freely available for use under the CC-BY-4.0 license.

Why a new Department?

1. Cell engineering has become a quantitative discipline.

Pan-organism engineering methods. Historically, working with a particular organism was determined by with whom you trained. This has changed as labs now work across cell types with ease, sometimes from bacteria through human cells or plants, empowered by a new toolbox of powerful, organism-agnostic design tools. Engineers in the workforce report having to change species smoothly when changing jobs inside or across companies.

Biophysical principles drive design. In the past, metabolic simulations, protein design, and bioinformatics could only be used to describe a natural system. While "mathematical biology" has been around since the 1970s, it became relevant to genetic engineering in the early 2000's when researchers could create virtually any DNA sequence (soon any genome). The increased reliance on mathematics, physical principles, and computation has turned the manipulation of living cells into a true engineering discipline.

2. We must expand the workforce by attracting more students and accelerating their training.

New industries are embracing biotechnology. Biotechnology once meant pharmaceuticals, food, and agriculture. Today, it includes chemicals, information technology, construction, energy, materials, architecture, mining, defense and aerospace. It has been estimated that up to 60% of the "physical inputs to the global economy could, in principle, be produced biologically."[1]

A growing need for a trained workforce. Biotechnology and genetic engineering produce attractive jobs; in 2023, the average U.S. biotech worker earned $132,000/year—83% above the national average.[11] About 2.3 million employees work at 150,000 biotech companies, and the sector grew 4% annually from 2021 to 2023.[15] Government calls to boost biotech training include Congressional Acts and Presidential Executive Orders mandating "multi-disciplinary education programs."[3-10]

The time to enter the workforce needs to be shortened. Biotechnology programs often require a PhD and postdoc for

students to become employable, leading many students to enter their first job in their early 30s. Other engineering disciplines can enter companies with a BSE. This delay is unnecessary; it not only extends the training time of those in the field, but also discourages others from entering it.[2] Note that we do not seek to eliminate PhD or postdoctoral programs. They remain critical for those wanting to enter scientist tracks in industry or to run academic labs.

> "The pace of technological advances and integration of tech and data is so rapid, if you spend 4 years to get a B.S., 5 years for a PhD, then 3 years for a postdoc, you will be specializing in things that will be quickly outdated."
>
> *Daniel Grenfell-Lee, DSM-Firmenich*

3. The bioeconomy is a global priority, and the U.S. risks losing its lead.

Biotechnology is a national and global priority. The future bioeconomy promises food security, sustainable economic growth, robust supply chains, and next-generation medicines. There is a global race to grab the potential economic advantage. In the U.S., there is bipartisan support for biotechnology. The 2012 National Bioeconomy Blueprint sought to "realize the full potential of the U.S. bioeconomy."[6] A 2019 Executive Order sought to enhance and deregulate biotechnology and genetic engineering, particularly in agriculture.[5] In 2022, Executive Order 14081 advanced "domestic biomanufacturing and biotechnology which will provide an alternative to petroleum-based production for chemicals, medicines, fuels, materials, and more."[3] Genetic engineering is also being prioritized elsewhere, notably in China, the European Union (Denmark, the United Kingdom, Germany), Turkey and Israel. Experts cite "inadequate talent development" as a leading reason the U.S. could lose its edge.[6]

Massive investments in genetic engineering. U.S. biotechnology

generates $3.2 trillion annually (5.1% of GDP), yielding $136 billion in state/local taxes and $216 billion in federal taxes.[2,11] Less than half of these investments are in medical applications. Globally, experts estimate that the bioeconomy could reach $30 trillion within two decades.[1] Venture investment in genetic engineering startups hit $7 billion in 2024. Large-scale efforts at DoE and DoD have gone to establishing a national shared resources for biomanufacturing housed at universities with injections of hundreds of millions of dollars. However, to maintain a lead over China, the funding for R&D and STEM fields would have to double relative to GDP over the next 5 years.[12-14]



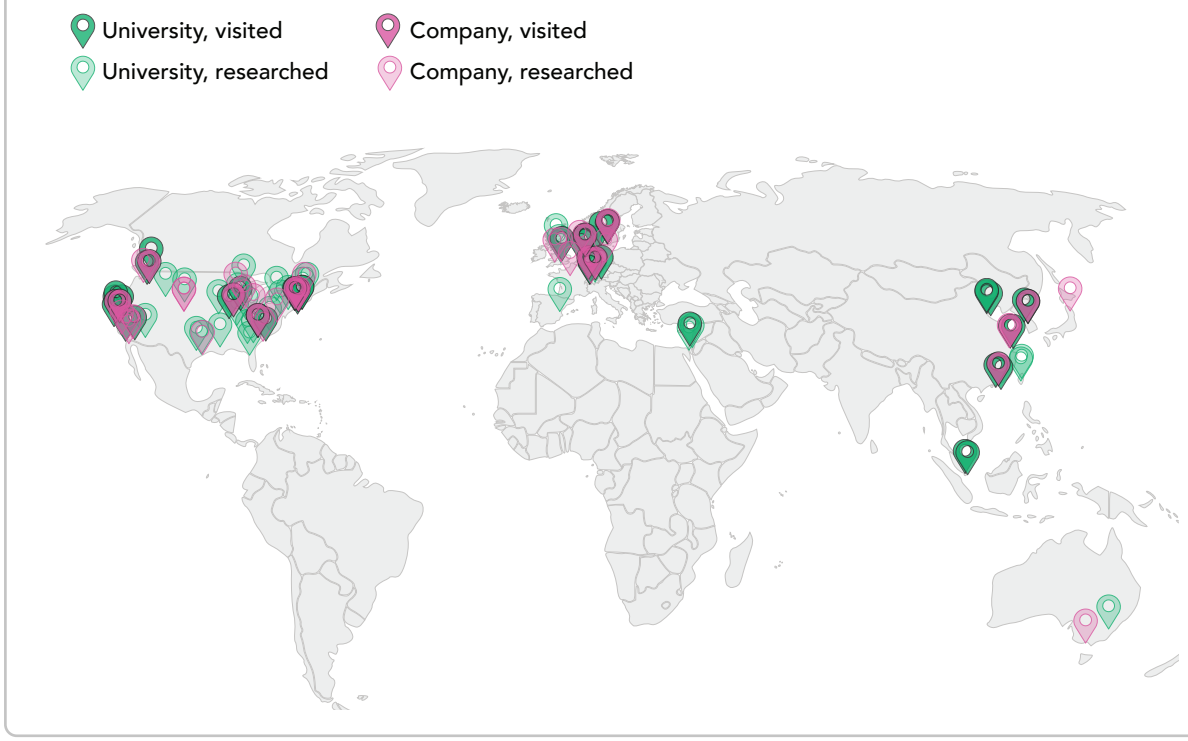
1. *Michael Chui, Matthias Evers, and James Manyika, et al., (2020) The Bio Revolution: Innovations Transforming Economies, Societies, and Our Lives, McKinsey Global Institute.*
2. *The U.S. Bioeconomy: Charting a Course for a Resilient and Competitive Future, A Bioeconomy Strategy. Schmidt Futures, April 2022*
3. *Executive Order on Advancing Biotechnology and Biomanufacturing Innovation for a Sustainable, Safe, and Secure American Bioeconomy, Sept 2022*
4. *Fact Sheet: Biden-Harris Administration's Actions to Advance American Biotechnology and Biomanufacturing, OSTP, June 2024*
5. *Executive Order on Modernizing the Regulatory Framework for Agricultural Biotechnology Products, White House, June 2019*
6. *White House, National Bioeconomy Blueprint, Washington, DC, April 2012*
7. *Office of Science and Technology Policy, Summary of the 2019 White House Summit on America's Bioeconomy, Washington, DC, October 7, 2019*
8. *Bioeconomy Research and Development Act of 2020/21*
9. *Engineering Biology Research and Development Act of 2019*
10. *The Bioeconomy: A Primer, Congressional Research Service Report, R46881, 2022*
11. *The U.S. Bioscience Economy: Driving Economic Growth and Opportunity in States and Regions, Bio, 2024*
12. *People's Republic of China State Council, Circular of the State Council on Issuing the National 13th Five-Year Plan for the Development of Strategic translation by the Center for Security and Emerging Technology, November 2016*
13. *Science & Technology Action Committee, China is a Determined and Formidable Competitor with the U.S. in Science & Technology.*
14. *Nazish, J. (2024) Bold Goals Require Bold Funding Levels. The FY25 Requests for the U.S. Bioeconomy Fall Short, Federation of American Scientists*
15. *U.S. Life Science Talent Trends, CBRE Research, 2024*

Global Programs Analyzed

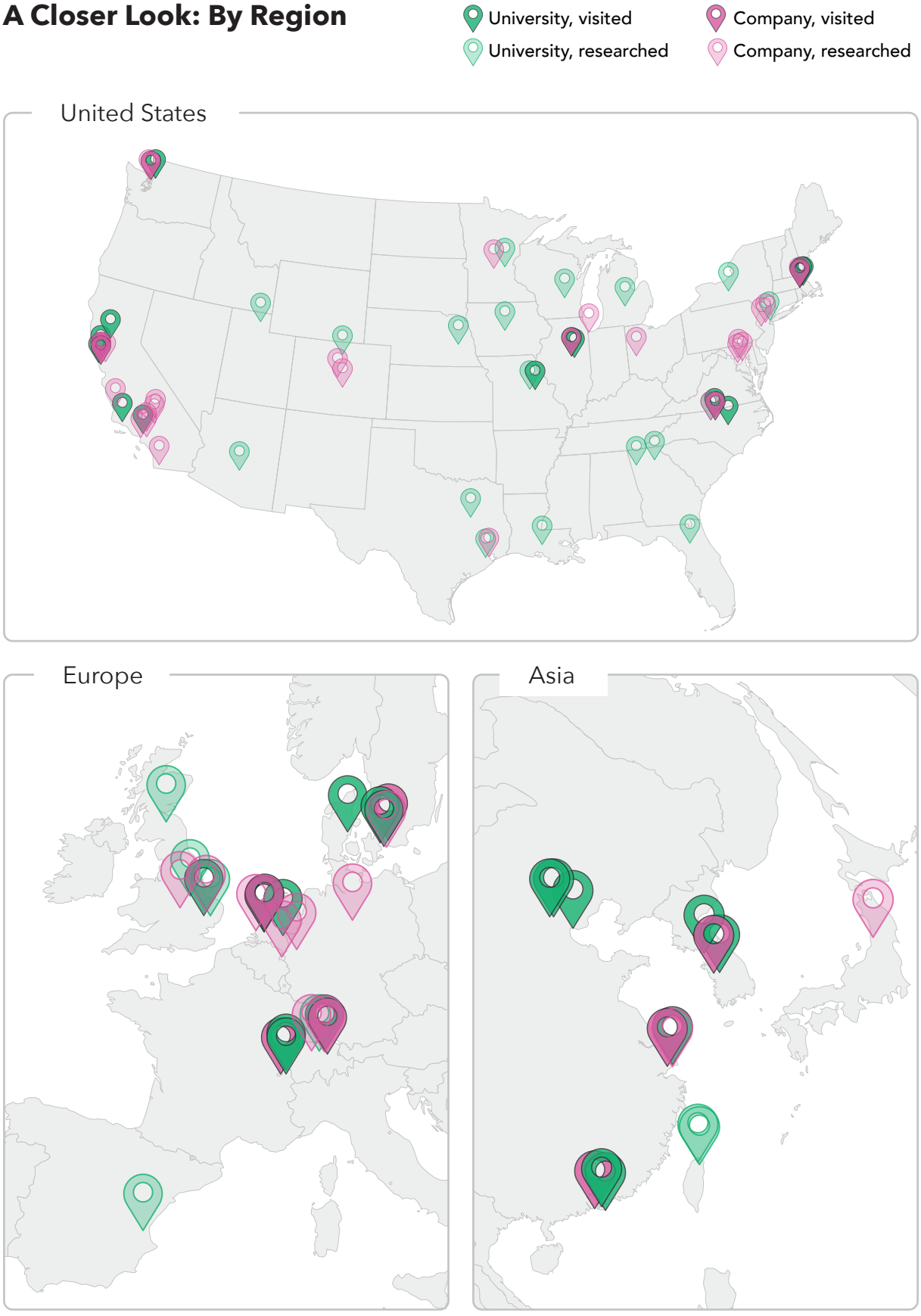
Genetic engineering research is housed in diverse departments worldwide. We visited 51 universities and research centers in North America, Europe, the Middle East, and Asia. We also obtained 90 curricula from 60 universities. Universities were selected either because they have a top-ranked program in different fields or by identifying concentrations of faculty or coursework focusing on genetic engineering.

We also interviewed employees from more than 100 companies across agriculture, therapeutics, chemicals, consumer products, defense, diagnostics, food and ingredients, materials, mining and pharmaceuticals (see Appendix). Companies also spanned seed-stage start-ups with a few employees to large-cap industrials with 10,000s; interviewed employees include everything from Research Associates to CEOs. Interview questions focused on the skills they seek in new hires, the future of genetic engineering, and how to build a relevant academic program.

Universities and Companies Analyzed



A Closer Look: By Region



For a full list of universities and companies analyzed, see pg 47 of the Appendix.

Where is Genetic Engineering Education now?

Genetic engineering research spans many departments across engineering and science. We visited and collected syllabi from global programs, focusing on those either highly ranked or notable for their work in genetic engineering.

- **Bioengineering - Medical.** Thousands of Biomedical Engineering departments exist worldwide. Their mission is to apply engineering principles to medical problems and are often partnered with a hospital. These programs have few courses or faculty focused on genetic engineering. Even coverage of basic biology relevant to genetic engineering is low. Most coursework emphasizes medical applications of engineering, featuring classes such as "Modeling Human Physiology" (University of Illinois), and a "Device Diagnostics and Design Lab" (Boston University). Curricula across these universities are very similar, partly because the field was established 50 years ago, is accredited by ABET (Accreditation Board for Engineering and Technology) and coordinated through the BioMedical Engineering Society (BMES).

- **Bioengineering – Molecular.** Rather than focusing on medical engineering, some bioengineering departments treat biology as the engineering substrate. Only three universities—Caltech, MIT, and UC-Berkeley—fully belong to this category, while Stanford and Johns Hopkins blend elements of both. However, only 1-2 classes are focused on genetic engineering. Because these departments are relatively new, they have no common accreditation or central society, causing wide variations in course topics. Most applications remain medical.

- **Biotechnology.** These programs are prevalent in Europe and Asia. They usually are not standalone departments, instead combining professors from different departments.

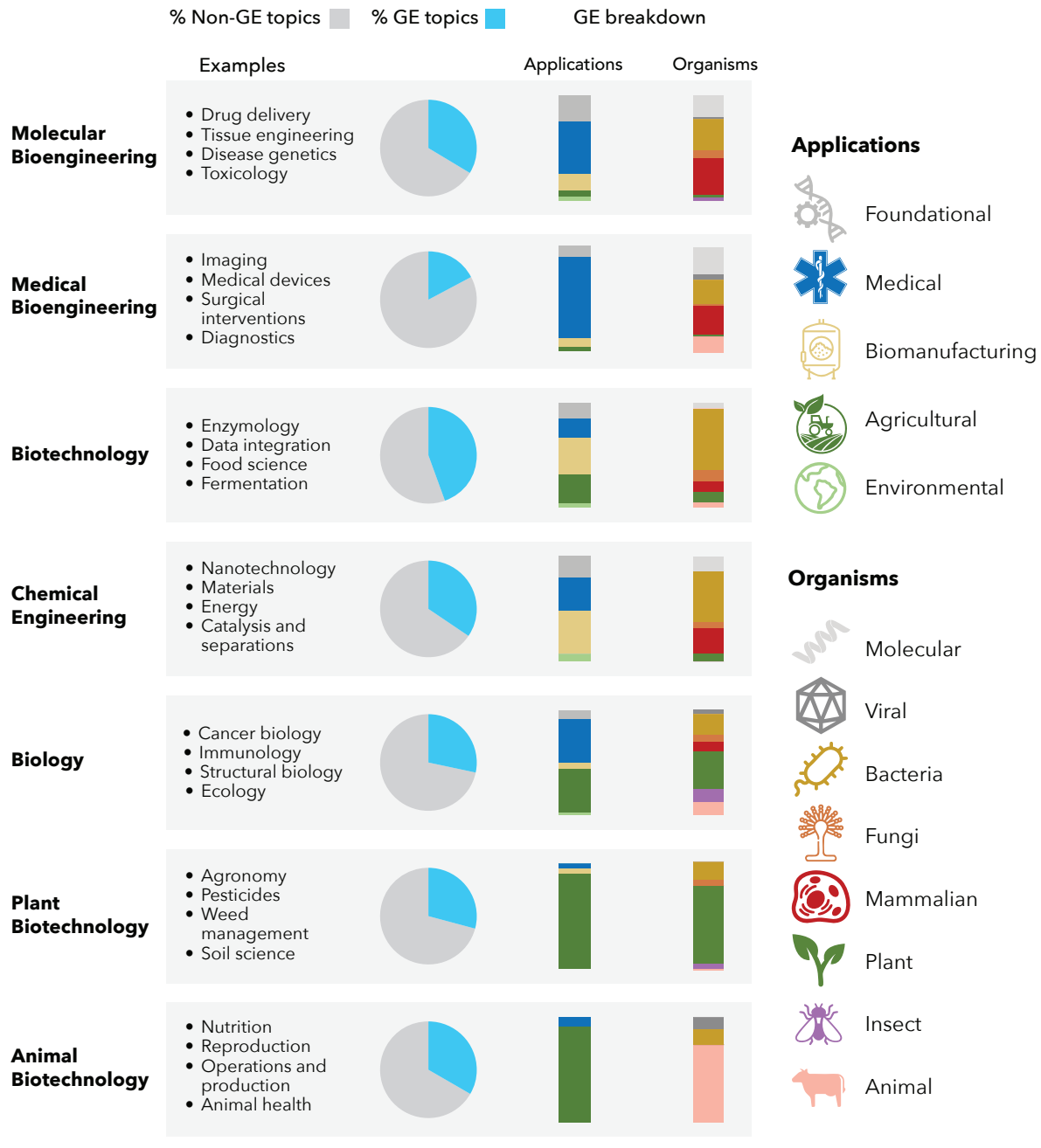
The curricula vary greatly depending on whom oversees it. In biology-led programs, mathematics and the physical sciences are often absent, whereas these subjects may predominate in engineering-led programs with little basic biology. Some curricula specialize in particular areas, such as fermentation, or focus on certain organisms, like algae. Of all the programs we studied, Shanghai Jiao Tong University's Biotechnology curriculum most closely resembles the one we envision (see page 21).

- **Synthetic Biology.** In Asia, multiple new "Departments of Synthetic Biology" are taking shape, with Tianjin University leading the way and similar models appearing across China. While the curriculum is relevant, its focus is on modern techniques, rather than underlying principles. For example, the curriculum includes courses on "Gene Circuit Design," "Gene Editing," and "Artificial Intelligence Biological Design."

- **Other disciplines.** The remaining departments in our analysis have a consistent footprint in genetic engineering through both coursework and faculty research. Although most Chemical Engineering departments include only a small fraction of biotechnology courses, some stand out, including MIT's Course 10B, KAIST, and Colorado State.

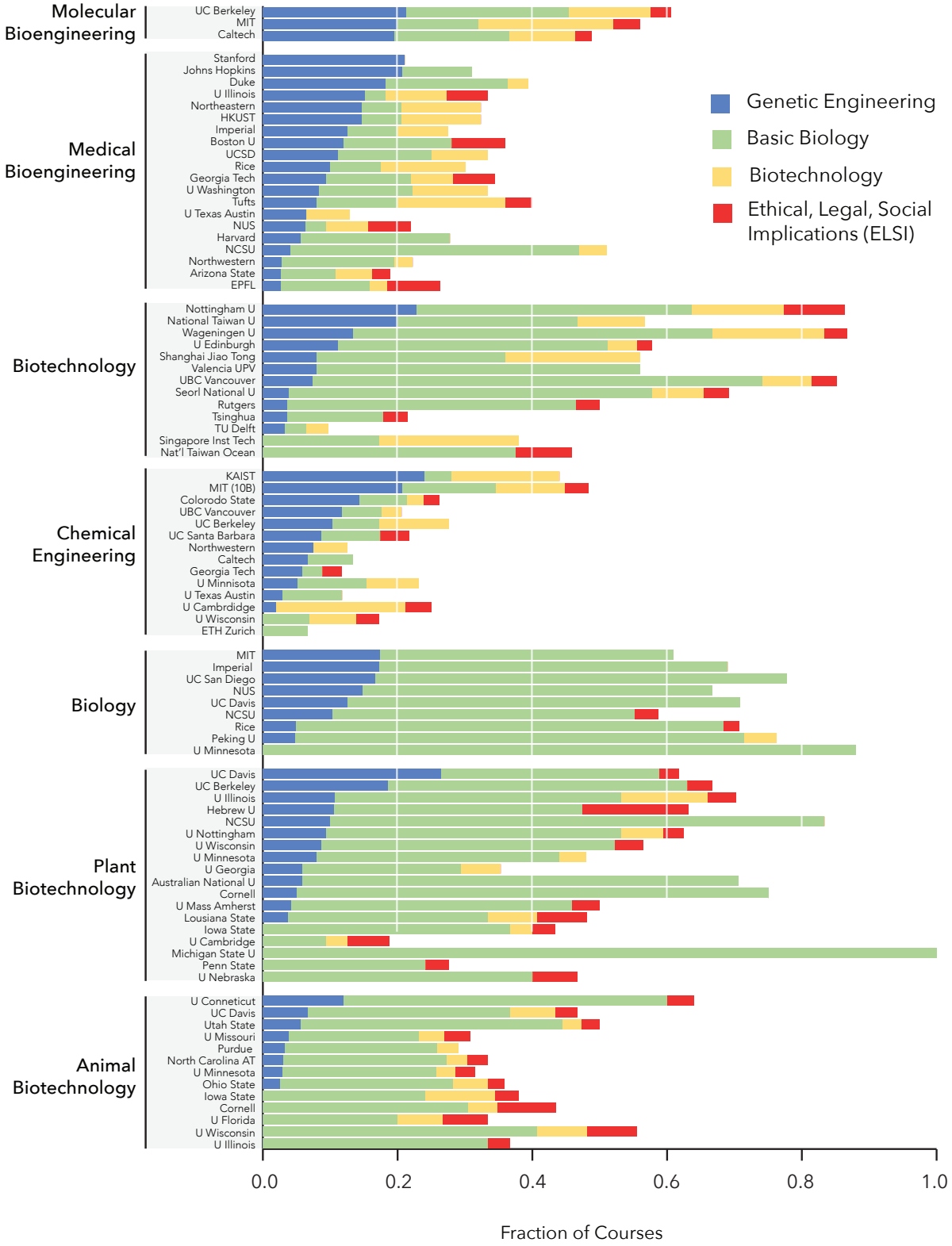
  Animal Biotechnology and Plant Biotechnology degree programs appear under different names, but almost all focus exclusively on agriculture. They include a few classes and faculty that work on genetic engineering. Certain Plant Biotechnology programs concentrate on specific crops or broader categories like grasses, trees, or ornamental plants. Similarly, some Animal Biotechnology programs focus on certain livestock types. Plant Biotechnology curricula often feature classes like "Soil Conservation and Land Use". Meanwhile, Animal Biotechnology programs may include classes like "Applied Equine Reproduction".

Faculty interests across departments



Data obtained from the analysis of the top 3 U.S. and international programs in each department type (Appendix II and III). The "top" programs were defined as those that teach the highest fractions of GE classes. "Non-GE" topics encompass the research areas that dominate these fields that do not focus on genetic engineering.

Distribution of classes across departments



The fraction of classes in each department type from 90 U.S. and international universities (Appendix II and III). When multiple tracks were available, we selected one most relevant to genetic engineering.



Biotechnology Program, Shanghai Jiao Tong University

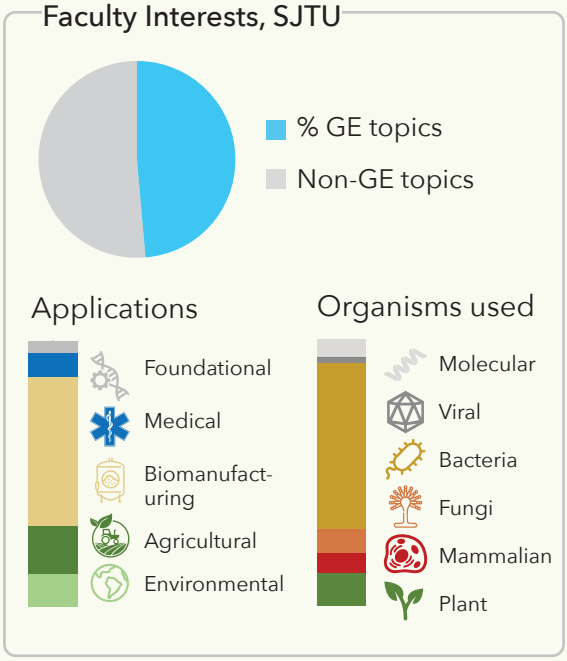
Founded in 1896, Shanghai Jiao Tong University is one of China's most prestigious universities. According to the 2024–25 U.S. News & World Report, its Biotechnology and Biomedical Engineering programs rank 8th worldwide. The university has built a modern Biotechnology undergraduate program that that is distinct from its peers and was the closest we observed to our vision. The program is supported by a diverse faculty composition and state-of-the-art teaching facilities.

The program's strength lies in its large portion of faculty focused on genetic engineering across diverse applications. In the Life Sciences and Biotechnology division, 178 faculty members work in Biochemistry and Molecular Biology, Genetics and Developmental Science, Microbiology, Bioinformatics and Biostatistics, Bioengineering, and Bio-X. Half of these faculty conduct research directly related to genetic engineering—one of the highest concentrations we observed anywhere. Their scope of organisms and application areas is broader than at most universities. A majority (72%) of these researchers work with eukaryotic microbes and bacteria, reflecting a concentration on microbes that produce natural products. Over half focus on biomanufacturing pharmaceuticals and other chemicals. Some researchers also engineer plants and insects for agriculture, and others study human cell lines for medical applications.

The undergraduate curriculum has a strong basis in mathematics, quantitative sciences, and computation. It also offers comprehensive basic biology coverage, with only a few courses considered less relevant, and most classes include both traditional lectures and hands-on lab work. Students can choose from more genetic engineering classes than are typical elsewhere, including biomolecular and cell engineering courses spanning multiple scales. They maintain a five-story building dedicated entirely to teaching labs, featuring standalone classrooms for fermentation, general biotechnology, and roughly a dozen other courses. Undergraduates gain hands-on experience in gene editing, protein design, and cloning.







Industrial Needs

The majority of undergraduate students with a Genetic Engineering degree would be hired by companies. To determine needs, we interviewed companies, focusing on future needs rather than immediate hiring gaps. We aim to graduate the genetic engineers of 2030 and beyond.

Vuewpoints

- **Engineer Versus Scientist.** Most engineering disciplines have two career tracks at companies. With a B.S.E., graduates can pursue an engineering track that involves practical applications of engineering principles. They begin as entry-level engineers, progress to Chief Engineer, and eventually move into management roles. In contrast, scientists usually need PhDs, begin as Research Scientists, move on to Principal Investigator roles, and can become a R&D directors or Chief Scientific Officers.

- **There is no engineer career track for a biological engineer.** Current graduates are ill-trained to practice genetic engineering with a B.S.E., requiring skills from a PhD and postdoc. Most of the companies we interviewed hire PhDs as Research Associates (RAs), who then advance Scientist I/II, and beyond. Even supportive companies assumed that a candidate with a BS in Genetic Engineering would start as an RA and compete with PhDs in that role. This gap will be addressed by restructuring the curriculum.

- **Genetic engineers are highly trained designers of living systems, not technicians to build DNA.** Many companies assumed that a Genetic Engineer's primary role would involve assisting in DNA construction or performing assays for a PhD-level RA. This bias stems from the view that "Synthetic Biology" focuses more on DNA construction, rather than on underlying design principles. This bias is less prevalent at companies where the product is an engineered living cell (e.g., cell therapeutics), or when the industry is new to biotechnology (e.g., mining). It is greater in industries that utilize larger numbers of scientists, like pharma, and where the genetic engineering tasks are relatively simple and compartmentalized.

Skillsets important to industry

- **Flexibility in working across organisms.** Flexibility in working across organisms would help graduates transition to different companies during their career. The ability to work with "non-model" organisms is sought after; particularly isolates (human microbiome, soil, anaerobes, etc.), filamentous fungi, and primary human cells.

- **Mathematics and computation-guided genetic design.** Predictions need to be quantitative and actionable; in other words, map to a change in the DNA.

- **Strong molecular biology core.** There needs to be a basic understanding of the underlying methods; for example, enzymology.

- **Next-generation sequencing (NGS) is undertaught and essential.** The only methodology that came up across sectors is a need for NGS analysis training. It supports soil/human metagenomics, disease SNP identification, finding natural product gene clusters, off-target CRISPR quantitation, and so on.

- **Statistics are essential and a new approach to teaching this subject is needed.** Nearly every company noted that students were ill-prepared to perform statistical analyses. This observation is interesting, because most undergraduate programs do teach statistics, indicating that the wrong approach is being taken. This may reflect that the type of statistics differs depending on application (field trials, clinical trials, genomics analysis, enzyme assays, high-throughput screening, AI training on datasets, design-of-experiments for fermentation, etc.).

- **Genetic engineering requires ELSI (Ethics, legal, Societal Issues) education.** Genetic engineering is one of the most controversial and impactful technologies developed by humans. Formal ELSI training is required, including bioethics, intellectual property, EPA/USDA/FDA regulation, GMP/GLP/ISO biomanufacturing, and clinical trials.

A New Genetic Engineering Undergraduate Program

We developed a course plan for a Genetic Engineering undergraduate degree that covers the design and construction of biological systems, from molecules to organisms. Quantitative biophysics and mathematics lie at the core of the design process, with computational support. Students compare genetic encoding across species. Laboratories increase in complexity, progressing from the design and analysis of single biomolecules to designing changes in complete plants and animals. ELSI courses appear in the first and last years of the program, with their concepts embedded in every class.

Keep in mind that this is not a technical degree. Therefore, the focus does not lie on modern genetic engineering methods or how to use equipment. No undergraduate class explicitly teaches CRISPR genome editing, directed evolution, or artificial intelligence. Instead, we emphasize the fundamental concepts that empower students to comprehend these approaches and eventually develop the next generations of techniques. Students learn principles and mathematical modeling linked to changes in DNA, leaving its physical assembly to whomever performs that construction in the future.

> “MIT’s computer science trains students to design the NEXT SET of tools. They do not learn the “latest tools;” they learn how to think. Students do not learn language of the day; they were learning the deeper principles.”
>
> *Jake Beal, Raytheon BBN*

Laboratory courses allow students to design GMOs, followed by physical construction. Early labs center on molecular and simple organisms, and later labs address building a GMO animal (insect) or plant. The senior capstone course involves designing a genome using computational tools for biomolecular, metabolic, and whole-cell simulations.

Note that this program does not aim to replace any other department. Existing departments perform research and teach areas crucial for implementing genetically engineered solutions in society. For example, GMO crops still require agronomy and soil science.

> "The genetic engineers in 7 years should be tool agnostic and think broadly about how to tackle various problems and find appropriate tools."
>
> *Alexi Goranov, SciFi Foods*



Undergraduate Curriculum Overview

		Design	Computation	Lab	ELSI	External
Sophomore Molecular	**Fall**	Intro to Genetic Design			Ethics of Genetic Engineering	Differential Equations
	Spring	Biomolecular Physics		Quantitative Biochemistry		
Junior Cellular	**Fall**	Metabolic Engineering	Intracellular Dynamics			Microbiology
	Spring		Cellular Composition	Genetic Engineering		Cell Biology
Senior Organism	**Fall**	Organism Design I: Plants Organism Design II: Animals			Rules and Laws of Genetic Engineering	Physiology *or* Plant Biology
	Spring	Capstone Project		Organism Construction		

Sophomore (Year 2)

1st Semester

- **Intro to Genetic Design**
  This course offers a comprehensive introduction to genetic engineering. Students will learn the fundamentals of how cells function, focusing on their metabolism, regulatory mechanisms, and resultant actions. The course emphasizes a simplified understanding of the mathematics and biophysics that underpin these principles. A key element of the course is developing a quantitative understanding of the scale and timing of biological processes. From the beginning, students will also learn how to connect theoretical models with specific DNA design choices.

- **Ethics of Genetic Engineering**
  This course develops an ethical foundation for the use of genetic engineering. It will cover historical developments and the ethical debates they sparked. Examples include:

  1. The 1975 Asilomar Conference on Recombinant DNA.
  2. Early insulin production using *E. coli* in high-security labs.
  3. The first environmental release of engineered bacteria (Frostban).
  4. The development of GMO crops, such as Roundup-ready varieties.
  5. Reconstituting the 1918 flu virus.
  6. Human germline editing with CRISPR.

  The course will present global perspectives and ethical standards through these case studies. Coursework will be designed around “open ended problems” involving professional dilemmas, requiring the application of guiding principles.

2nd Semester

- **Biomolecular Physics**
  Students will learn how to design biomolecules, from chemicals (metabolites) to RNA and proteins. Principles will be taken from thermodynamics to calculate molecular interactions and self-organization, including ligand design, lipid self-assembly, and protein/mRNA folding. Problem-solving exercises will highlight the use of fundamental theories in genetic design to achieve specific functions in biomolecules or DNA sequences. Computational tools will link fundamental concepts to the design of biomolecules to create structures and enzymes not seen in the natural world.

- **Quantitative Biochemistry Lab**
  Enzymes are the core of all biological systems. This lab encompasses work central to modern, quantitative enzymology. Theory will be presented on the derivation of enzyme kinetics. Advanced topics will include diffusion-limited catalysis, chaperones, co-factors, and single-molecule enzymology. Laboratory projects will avoid the use of outside reagents and kits. Example labs include:

  1. *Enzyme mining*. Identify and characterize an enzyme in the genome of a plant.
  2. *Single molecule enzymology*. Measure the activity of individual enzymes using microscopy.
  3. *De novo enzyme design*. Design a sequence predicted to fold into an assigned target three-dimensional structure and place amino acids for simple catalysis.
  4. *Drug discovery*. Use docking software to predict compounds that will inhibit an enzyme from a set of provided compounds. Measure binding and inhibition kinetics.

Junior (Year 3)

1st Semester

- **Metabolic Engineering**
  Metabolism is the complex network of chemical conversions catalyzed by enzymes, and its manipulation can lead to new drugs, chemicals, and fuels. This course will describe mathematical modeling of metabolism and cellular energy, focusing on differential equations and linear algebra to solve genome-scale models. Mass action kinetics will be taught. The course will focus on teaching intuitions around timescales and number of molecules, as well as the breakdown of some macroscopic principles at small scales, such as pH. Practical biomanufacturing will be introduced, including theoretical yields and titers.

- **Intracellular Dynamics**
  Cells exist as densely packed, dynamic environments. This course will focus on physical phenomena at the scale of a cell. Topics include fluid dynamics and molecular diffusion in the cell, including by electrically-charged molecules. Emphasis will be placed on understanding when and how simplifying assumptions can be made. How molecular machines direct motion to surpass diffusion limits on large molecules and organelles will be discussed as well as the thresholds at which small molecule numbers necessitate stochastic modeling. Intracellular phase changes in chromosome packaging and their movement due to microtubules will be modeled. Design will be introduced with a whole-cell simulator of a model bacterium.

2nd Semester

Cellular Composition

Cells are a complex milieu of nucleotides, metabolites, lipids, and proteins. If you cannot "see" the effects of genetic manipulations on this composition, it is hard to debug or guide the cellular design process. Students will explore techniques to evaluate the molecular composition of molecules in the cell. The course will be taught in four parts to describe modern techniques in cellular imaging, and then the composition of mRNA, proteins, and metabolites. This is a lecture-based course, focusing on the processing of data generated by different techniques. Computational methods and underlying algorithms for processing large data sets and image analysis will be taught.

Genetic Engineering Lab

Students will design and characterize synthetic living cells. Lectures will focus on the theoretical underpinnings of each computational tool used. Labs will focus on the design of a cell or biomolecule to make or do something not seen in nature. The class will emphasize electronic data recording practices, applied statistics, and communication.

- *Pathway design for a new-to-nature molecule*. Computational retrosynthetic design of a pathway to make an assigned target molecule. Sequences will be selected from a limited set of enzyme sequences known to be active in a model organism. The molecule will produce a color change that can be assayed by spectroscopic methods.

- *Redirect flux to desired metabolite*. Apply metabolic engineering tools to direct genome changes that increase the amount of sugar converted into a target selected that can be easily assayed, such as smell, color, or antimicrobial activity.

Senior (Year 4)

1st Semester

- **Organism Design I: Plants**
  This course teaches plant physiology and genetics from the perspective of an engineer. The course outlines genetic engineering of nuclear, chloroplast, and mitochondrial genomes, plus metabolic pathways that span these organelles. It also teaches considerations when manipulating monocot/dicot and C3/C4 species. Methods to control plant development and morphology are taught, as are root-to-shoot signaling, plant defense and stress response mechanisms.

- **Organism Design II: Animals**
  This course teaches animal physiology from the perspective of an engineer, spanning applications across health, environment, and agriculture. The genetic engineering of animals will include topics such as promoter design, sensing and circuit design, and development. Design will span different cell types, such as immunological and neuronal cells. Modern applications will be covered, including the design of animals to humanize organs for transplant, control of livestock phenotypes, or insect engineering for ecological control.

- **Rules and Laws of Genetic Modification**
  Legal topics relevant to biotechnology and genetic engineering will be taught. Similarities between U.S. and global regulations will be covered, including the scope and responsibilities of national and state regulatory agencies. EPA, USDA, and FDA statutes, regulatory authority, and processes will be described. Regulatory differences between green, white, and red biotechnology as well as the approval of GMOs for contained versus environmental release. The basics of clinical trials will be compared. An overview of international treaties and law (Nagoya Protocols, United Nations Treaties) will be given. The basics of patent law will be covered, with an emphasis on proficiency in locating and reading patents/claims.

2nd Semester

- **Capstone Project**
  Students will design the genome of an organism (microbe, plant, or animal) to achieve a target task. Computational methods will be applied to DNA design and experiments for the physical construction will be planned. Depending on the project, models will be developed for the function desired, such as the determination of enzymatic changes to redirect flux, *de novo* enzyme design, retrosynthesis, genetic part calculations, and simulation of signaling or stress response. Simulation environments will evaluate the performance of their designs, which may include cellular all-molecule, growth, and developmental models. Communication and writing skills, as well as team-oriented science, will be emphasized.

- **Organism Construction Lab**
  This lab course will teach the physical construction of a plant or animal with desired traits. Design tools will be applied to create desired constructs, building on the junior-level Genetic Engineering Lab. Fruit flies will be used as a model of animal engineering, whose chromosomes will be modified to alter behavioral and visual phenotypes. Changes in the DNA sequence of the model plant Arabidopsis will be made and the effect on phenotype will be measured. This could be physiological cell growth and morphology (height, number of flowers, root morphology) or synthetic (light or color production). For both projects, quantitative modeling will be emphasized when directing DNA changes. Rigorous statistics will be integrated into the data collection and reporting.

Graduate Coursework

The mission of the department is to be the top PhD-granting program that practices genetic engineering. A core focus is on preparing those graduating with a BSE to pursue engineering careers in industry or government. Our philosophy for the role of a PhD program, however, is to train graduates prepared for scientist career tracks at companies or to run academic or government labs.

The goal of graduate coursework is different than its undergraduate counterpart. Graduate students would be admitted from many of the departments that perform some genetic engineering, such as bioengineering, chemical engineering, plant science, animal science, biology, and departments whose core curriculum is much different (mechanical engineering, physics, chemistry, etc.). Therefore, the principal objective is to get all of these students, coming from different backgrounds, to common intellectual ground. This task also has to be done with the minimal amount of class time because the main purpose of graduate school is to learn by performing original research in a laboratory.

Most of the incoming students will not be well educated in two of the department's pedagogical principles (page 3). First, "universal species design" encourages cross-species engineering, but many students arrive from departments focusing on a limited subset of organisms. The second is "mathematics-guided predictive engineering," where different genetic engineering departments have different levels of training in applied mathematics, physical sciences, and computational algorithms. Four courses will be taught, all to first year students as they are rotating and selecting a lab. These courses also differ from their undergraduate counterparts in that they tend towards advanced topics, and thus are closer to modern computational and experimental techniques that will likely impact their research.

1st Semester

- **Advanced Organism Design**
  This course teaches topics in mathematics and computation. There will be 3-4 modules that begin with topics in

biophysics, followed by advanced topics that apply those principles to genetic design. For example, a module could be on the biophysics of protein folding, followed by a modern technique (e.g., AlphaFold). Similarly, principles of metabolic engineering (linear algebra), regulatory design (non-linear dynamics, stochastic simulations), and NGS bioinformatics (statistics) will be taught. Unlike the undergraduate curriculum, the syllabus will be updated with modern techniques (generative AI, CRISPR, machine learning, etc.).

- **The Physiology of Living Things**
  The focus will be teaching physiology across all kingdoms of life. The first half will span all microbes, including archaea, bacteria, yeast, fungi, and algae. The second half will cover multicellular organisms, including the process of differentiation, taxis, and specialization.

2nd Semester

- **Quantitative Cell Biology**
  The focus will be on developing quantitative intuition regarding cellular processes. Material will be presented as physically-meaningful numbers (number of a protein per cell, free energies of interactions, etc.). Second, topics will be taught across the kingdoms of life. For example, for the central dogma, the topic of transcription will span phage/bacteria, eukaryotic microbes, and human/plant/insect cells. The topics will include genome organization, DNA topology and modification, transcription and translation, protein-folding and post-translational medication, membrane processes, metabolism, and organelles. Emphasis will be placed on how changes to the DNA sequence, as might be enacted by an engineer, affect these processes.

- **Advanced ELSI topics**
  This course will present four modules. The first module will be on ethics, focusing on the ethical code developed by the department and other stakeholders. The second is on intellectual property and patent law, including how to interpret patents and search the patent databases. The third will be on regulatory frameworks, such as the permitting and approval processes across application spaces. Finally, the basics of business development will be taught, including the entrepreneurial steps to starting a company and raising funds.

Ethical Frameworks

Despite a half-century of research, there is no formal ethical code for Genetic Engineers. After the first gene transfer experiments, a conference was held at Asilomar in 1975 to determine the guidelines for safe pursuit of experiments with recombinant DNA. To reduce risks to human health, they established biosafety levels (BSL) 1-4 and established institutional biosafety committees (IBCs), both of which are in widespread use globally. These are detailed and updated in the "NIH Guidelines for Research Involving Recombinant or Synthetic Nucleic Acid Molecules," which provides extensive risk mitigation and regulatory strategies for different classes of experiments, but not an ethical code.

Eventually, many of the guidelines limiting genetic engineering research fell by the wayside. For environmental safety, scientists recommended weakened strains and unstable plasmids. They banned certain experiments, such as enhancing pathogen virulence through "gain-of-function" research, exchanging genes among species, and editing the human germ line. Researchers were urged to self-regulate and weigh ethical concerns, but no department or society built explicit frameworks.

Genetic engineering represents one of the most controversial areas of research, with ever more powerful capabilities pushing boundaries. Some of the first engineered bacteria, including the first patented bacterium, were designed for bioremediation of pollution in the 1970s, but public outcry has kept this application from being implemented. GMO crops, originally developed by Monsanto in the 1980s, remain contentious and are banned in large parts of the world. Food and consumer products continue to be stamped "non-GMO" in an attempt to increase sales. The reconstruction of the influenza virus 1918 pandemic via DNA synthesis led to a robust public debate on gain-of-function research that continued through the COVID19 pandemic. In the 2000s, sugar-based fuel production sparked a "food versus fuel" dispute. Controversial issues have multiplied since the birth of CRISPR-edited twins (2019), advanced microbe engineering for environmental release (EMERs), and the Department of Defense citing biotechnology as a strategic priority. Looking ahead,

> "In a 2017 study, the [National Academies] recommended that federal agencies invest in new methods of understanding the ethical, legal, and societal implications (ELSI) of future biotechnology products. NASEM noted that ELSI research associated with biotechnology represented less than 1% of the total investment in biotechnology research between 2008 and 2015."
>
> *Preparing for Future Products of Biotechnology, The National Academies Press*

potential controversies include complete synthesis of human chromosomes, genetically tailored bioweapons, and "mirror" bacteria.

Several ethical codes exist for allied fields, but they do not map well to genetic engineering research. They are typically domain-specific and consider only certain applications. The BMES (BioMedical Engineering Society) maintains one of the closest existing codes, emphasizing medical and public health ethics. Although some tenets match Genetic Engineering's requirements, most do not apply, and the relevant sections lack sufficient detail. The Engineering Biology Research Center (EBRC) is a promising place to develop and house ethical guidelines; as it stands, their ethical code is too generic to be helpful in guiding research. Some codes have relevant points, but they do not have sufficient points to encompass genetic engineering (e.g., AiChE). We were unable to find ethical codes for the engineering of insects, plants, or animals (except those used for medical research).

Developing such an ethical code will require coordination across like-minded academic programs, as well as other stakeholders. These include industry, the public, and those that could be impacted by the technology.

Facilities to Enable Integrative Genetic Engineering

The envisioned Department would require capabilities to perform a unique combination of research projects. A challenge is that desire to work with many organism types could lead to a never-ending need for bespoke spaces. To understand what is required, more than 50 facilities around the world were visited. Beyond research, we looked carefully at educational spaces and attached incubators to translate research to society. Some observations were surprising and led us to partner with the architectural firm Ellenzweig to devise a new approach for organizing academic genetic engineering research.

These observations were reduced to **three architectural tensions**:

1. Balancing flexible lab space without creating impersonal "infinite" bench spaces (see Box).

- The most successful laboratories feel lived-in, but without clutter.
- Central communal spaces, such as a kitchen, promote community.
- Routine experimental tasks do not need to be done at the same location every day.
- Movement to large-scale projects requires common equipment space to be easily reconfigured.

2. Need for flexible spaces that can be reconfigured for most organisms, without an ever-expanding space requirement.

- Diverse viruses, microbes, and invertebrate animals have surprisingly similar space needs.
- Plant engineering needs additional features, including greenhouse; locating these far away hinders research.
- Vertebrate embryo manipulation can occur in the building, but more extensive vertebrate work often calls for specialized facilities maintained at the institute level or in partnership with biomedical or agricultural programs.

3. Need for flexible spaces that can be reconfigured for most organisms, without an ever-expanding space requirement.

- The Rosalind Franklin Center in Delft was used as a model to address some of these issues (see Box). That facility is industrial and focused only on microbial engineering; we took the best aspects of it, expanded the organisms that could be engineered, and adapted it to an academic setting.

Current Extremes of Biotechnology Space

Typical academic labs are organized by the PI and involve closed-off, locked labs. This organization limits flexibility, as it becomes a major and politically-charged effort to add or remove space to accommodate changes in research projects. It also negatively impacts collaboration and a sense of community. To address this problem, starting in about 2000, large shared labs began to be designed. Because molecular biology often demands the same basic setups, many labs merged into single open spaces. Though this approach could solve the problem, these spaces often seem starkly empty with only a few people working at a given time.

The MIT Department of Biological Engineering (left) and Synthetic Biology Center (right) are shown as examples of the extremes.



Introvert paradise



Infinite corridor of biotech depression

The Rosalind Franklin Center



This building was constructed in 2018 for the food ingredients and enzymes company DSM (150,000 ft²) in Delft, Netherlands. Prior to its construction, the company's research was performed in traditional facilities; people sat in assigned office locations and worked at specified benches. The Rosalind Franklin Center sought to upend this structure. By eliminating territorial spaces assigned to individuals, many more researchers can be fit into a smaller space and the building has a feeling of buzzing activity.

Open atrium. The public can access a small portion of the first floor, which includes some art and a view of the entire facility through the atrium. Because of this structure, visitors feel like they can view much of the activity in the building without entering it.

Central community areas. Kitchen, coffee, and socialization space are central to each floor. They are connected to a large atrium that runs through the center of the building. This creates an open feeling of community that spans floors.

Unallocated office space. Researchers store their computers and other office supplies in locked cubbies at night. In the morning, they can come and enter an open work area with long tables, which resembles a study area in a college library. Unassigned soundproof offices can be selected when reduced distraction is needed. Conference rooms of various sizes are available for meetings.

Flexible laboratory space. Researchers place their personal molecular biology reagents and personal equipment in a bin. This bin is stored overnight in a locker. When they want to do research, the bin is taken to an open bench. (Notably, the Bell Labs facility in New Jersey had the same design; each scientist was allocated about six feet of space, and the walls could be reconfigured as needed to reassemble teams on projects.)






Left to right: Central atrium, dry work spaces, unassigned benches, cores requiring more permanent housing

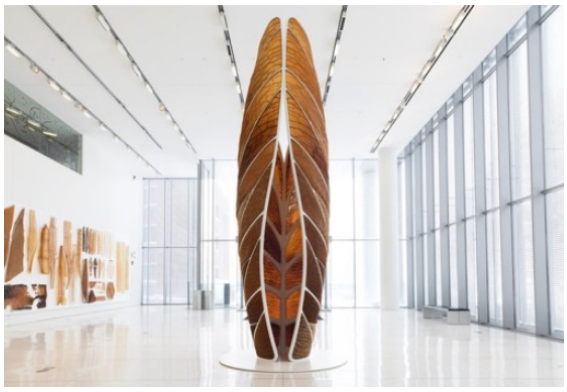
A Space for Public Engagement

The building will be open to the public and they should be given as much access and educational material as possible. This will be achieved through an open layout, offering windows into the labs, and monitors and exhibits explaining the research.

Left: The Museum of Postnatural History (Pittsburgh, PA) is the only resource for the collection of genetically modified organisms. They have acquired a collection of bacteria to animals that are shown globally.

Right: The atrium of the Media Lab at MIT showing an exhibit of engineered chitin by Neri Oxman. This area has rotating artistic exhibits displaying the work across the Lab.





Below: The atrium of the Broad Institute of Harvard and MIT, a large-scale sequencing center and research institute. The displays describe the benefits of genomic information, give real-time updates on projects (e.g., number of nucleotides sequenced), and show artifact equipment showing how sequencing has scaled up.



Architecture for a New Genetic Engineering Space

Ellenzweig, the same firm that created MIT's Koch Institute, designed the building. The design supports an estimated 40–50 principal investigators in a notably smaller area than standard academic labs. Most labs are open access. There are different types of rooms for working at computers or holding meetings that can be reserved. Same with laboratory space – students are not assigned a bench that they work at and a desk that they store all their stuff. Benches are first come first serve in the morning. Students would have "lockers" for their laptops and experimental supplies (media, Pipetman, etc.). Core to this approach would be a staff to run the facilities. The coldness common to biotechnology spaces was toned down through design elements, materials selection, and color tones. The exterior of the design is shown on the back cover.

Central lab rooms would focus on PI-specific equipment needs and shared equipment cores, such as analytical tools. Bespoke processes and lab storage could be in a room and larger equipment could be permanently situated in the community space. Routine work would be done outside of these rooms.

A large, open area allows for the placement of equipment. It is reconfigurable so that it can adapt to larger-scale, multi-investigator projects as needed. This equipment will be viewable from the public atrium, along with informational plaques and screens describing the research being observed.

Rooms are generically designed to be reconfigurable to different organisms, from phage to animal embryos.

They are designed to be small, contained, and have the necessary air pressure, water, and electrical power required for different organisms. Light rooms for plant experiments are in the basement and automated greenhouses are on the roof.

The public-facing view resembles a natural history museum, showcasing genetically modified organisms. The museum presents the material in an apolitical, frank tone; it simply shows what was done and leaves interpretations to the viewer.



The museum continues to the interior, providing a mixture of pieces acquired elsewhere with descriptions of the work being actively developed by the Department. Note the reliance on natural wood material and soft tones.

The museum opens onto a central atrium that offers views of both the upstairs laboratories and the large equipment or multi-investigator projects below. Informative displays, including electronic updates, keep visitors aware of ongoing research in these areas. This design provides a transparent look into the lab work, promoting greater understanding and public engagement.



Public viewing of the research area housing large equipment and multi-investigator initiatives.





The large, central area is reconfigurable for common equipment. The modularity allows for reconfiguration around project needs. The offset between the researchers and the public viewers avoids distraction.



Educational Lab Space. Desk designs promote team-based experimental work. Note the addition of drop-down utility lines to ease the reconfiguration of equipment at the bench for different labs.



View of the open laboratory and work spaces. (Foreground) Lab benches are available on a first-come first-serve basis, and researchers store personal equipment in the cubbies at the back. (Background) The open workspace contains unassigned benches where people can sit for dry work. A quiet room is shown on the left. (Middle) A central community area, accessed by stairs and an elevator, connects the wet and dry lab zones. Reservable conference rooms lie to the right. This floor would be closed to the public.



Large lecture hall for classes and conference facilities. Note the nautilus design.

APPENDIX

Pg. 47: List of universities and companies analyzed

Pg 48: Acknowledgements

Pg. 49: Glossary

Appendix II: Genetic Engineering Education and Research, an Analysis

- Undergraduate Curricula with Some Genetic Engineering Training
- Department Research Areas
- Training Needed Across Industries
- Facilities for Genetic Engineering Research and Education

Appendix III: Raw Data Supporting Analysis

- Undergraduate Curricula
- Faculty Research Areas
- Ethical Codes from Related Disciplines
- Genetic Engineering Facilities, additional photos
- Industry Interviews

Universities

Visited
California Institute of Technology
UC, Santa Barbara
UC, Berkeley
UC, Davis
North Carolina State University
University of Washington
University of British Columbia
Stanford University
Donald Danforth Plant Science Center
University of Illinois at Urbana-Champaign
Imperial College London
Aarhus University
University of Copenhagen
Harvard University
Novo Nordisk Center for Protein Biology
DTU Biosustain
Delft University of Technology
Leiden University
Wageningen University & Research
ETH Zurich
University of Lausanne
École Polytechnique Fédérale de Lausanne
Weizmann Institute of Science
Massachusetts Institute of Technology
Northeastern University
Boston University
KAIST
Seoul National University
Korea University
Peking University
Tsinghua University
Institute of Genetics and Developmental Biology, Chinese Academy of Sciences
Tianjin University
North Carolina A&T State University
Shanghai Jiao Tong University
Shenzhen Institute of Advanced Technology
Hong Kong University of Science and Technology
National University of Singapore
Singapore Institute of Technology

Researched
Rice University
Arizona State University
Tufts University
Johns Hopkins University
Swiss Federal Institute of Technology
Technion – Israel Institute of Technology
Duke University
Georgia Institute of Technology
University of Texas at Austin
University of Minnesota
Colorado State University
University of Cambridge
University of Edinburgh
National Taiwan Ocean University
University of Nottingham
Universitat Politècnica de València
Rutgers University
National Taiwan University
University of Georgia
UMass Amherst
Australian National University
Louisiana State University
Michigan State University
Cornell University
University of Wisconsin–Madison
Pennsylvania State University
University of Nebraska–Lincoln
Hebrew University of Jerusalem
Iowa State University
Purdue University
Utah State University
University of Connecticut
University of Missouri
University of Florida
Ohio State University

Companies

Visited
LatchBio
Asimov
Arzeda
Ginkgo Bioworks
Pivot Bio
Lawrence Berkeley National Laboratory
Pow.bio
Bio-Technopark Zurich
Biozentrum, University of Basel
Institute for Protein Design
RIPE, University of Illinois
Inari
KRIBB
BioInnovation Institute
NEXTDelft
BGI
BluePHA
Research Triangle Park

Researched
Amyris
Form Bio
Impossible Foods
DSM
Synlogic
Bolt Threads
LanzaTech
Mammoth Biosciences
Conagen
Codexis
Ecovative Design
Lattice Bio
Circe Bioscience
Biomason
Proteinea
Novome Bio
Invaio
Roche
Kite Pharma
Ensoma Therapeutics
Beam Therapeutics
Sangamo Therapeutics
Huue
New Culture
Sana Biotechnology
Tessera Therapeutics
Cataya Therapeutics
Catalog Technologies
Arbor Biotechnologies
TRIANA Biomedicines
BigHat Biosciences
Empress Therapeutics
Shiru
Factor Bioscience
Estée Lauder
Air Protein
Naval Research Laboratory
Aanika Biosciences
Inari
GenScript
NIST
Lockheed Martin
Air Force Research Laboratory
MITRE Corporation
Rio Tinto
Raytheon BBN Technologies
Moderna
Lygos
Twist Bioscience
Stemloop
Senti Biosciences
Finless Foods
Spiber
Life Edit Therapeutics
TeselaGen
Illumina
Sartorius
SciFi Foods
Oxford Nanopore Technologies
Robigo
Inscripta
Novo Nordisk
Cemvita
Eligo Biosciences
Manus Bio
Veramaris
Google Labs
Biocentis
Sampling Human
Syngenta
Cargill
GRO Biosciences
Perfect Day
Sound Agriculture
Pfizer
Provivi
National Resilience
Thermo Fisher Scientific
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Dan Mandell, GRO Biosciences
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Elliot Tague, Stealth startup
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Ernesto Simon, Cemvita Factory
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Gijs Verkleij, Spiber
Guillermo Montoya, Novo Nordisk
Heer Mehta, Cemvita Factory
Hui Zhou, Ginkgo
Huimin Zhao, UIUC
Ian Haydon, University of Washington
Ian Marshall, Aarhus University
Ian Wickersham, MIT
Ilya Tikhman, Pfizer
Iskra Katic, Friedrich Miescher Institute
Ismet Caglar Tanrikulu, Factor
Iulian Boboescu, Wageningen University
Jake Beal, Raytheon (BBN)
Jamie Richards, Aanika Bio
Janelle Vazquez, Impossible Foods
Jay Keasling, LBNL
Jeff Stuart, Lockheed Martin
Jefferson Smith, Oxford Nanopore
Jenna Usprech, UBC Vancouver
Jennifer Brophy, Stanford University
Jessica Sacher, Phage Directory
Jesus Fernandez, Eligo
Jin-Jie (Gogo) Liu, Tsinghua University
John Davis, Novo Nordisk
John Dileo, MITRE
Jonathan Chestnut, Thermo Fisher
Jorgen Kjems, Aarhus University
Josh Kittleson, New Culture
Julius Fredens, NUS
Julius Lucks, Northwestern University
Justin Beller, Lonza
Justin Buck, MIT
Justin Wang, Roche
Kaci Jenkins, Rio Tinto
Kangjian Qiao, Beam
Keld Lars Bak, Aarhus University
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Glossary

Genetic engineering (GE): The design of biological systems to alter their function or produce products.

Biological systems: Living and non-living biomolecules, viruses, cells, organisms, consortia, and ecosystems.

Biotechnology: The application of biological systems for industrial and other purposes.

Bioeconomy: The production and consumption of products, services, and processes derived from biological systems, spanning sectors including energy, agriculture, materials, chemicals, and medicine.

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McCarty, N., and Voigt, C.A. (2025) Department of Genetic Engineering: A Roadmap



A department in one building engineering all kingdoms of life.

"Biotechnology is working toward a world where diseases … are effectively treated, crops are highly productive and resilient, food production is more abundant and safer, industrial materials and chemicals are green and sustainable, and the environment and natural resources protected. [Its] innovations are the basis for dynamic new industry sectors with high paying jobs."

The U.S. Bioscience Economy: Driving Economic Growth and Opportunity in States and Regions, Bio, 2024

"Biotechnology will dramatically and literally reshape our lives and our world. It will also become a significant source of national power—economic, and in all likelihood military—as it creates entirely new possibilities, materials, and products. The question is whether our government can best position the United States to capitalize on this promise."

Dr. Tara O'Toole, Executive Vice President and Senior Fellow at In-Q-Tel and former Under Secretary for Science and Technology at the Department of Homeland Security, November 20, 2019

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